MIL-H-36809C
30 June 1987
SUPERSEDING
MIL-H-36809B
March 3, 1977

MILITARY SPECIFICATION

HANDPIECE, DENTAL, (Right Angle, Prophylaxis, Round Sleeve, Screw Stem)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an autoclavable right angle, prophylaxis, dental handpiece for attachment to a straight handpiece with a round nose, and for use with screw stem type polishing brushes and cups.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

GG-I-526	-	Instruments, Dental and Surgical; General Specification.
PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Methods of Preservation-Packaging.
MIL-M-7298	-	Manual, Technical, Commercial Equipment.
MIL-L-10547	-	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151	-	Metal, Test Methods.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging of.

DRAWING

Defense Personnel Support Center (DPSC)

24488 Handpiece, Dental, Right Angle, Prophylaxis, Round Sleeve, Screw in Type.

(Copies of specifications, standards, and drawings required by manufactures in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B-16 Specification for Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines.

E-18 Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

E-92 Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General Requirements. The components of the handpiece shall be in accordance with the applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Materials, Hardness and finishes. The handpiece components shall conform to the requirements as specified in Table I.

TABLE I. Materials, hardnesses and platings

PART NO.	COMPONENT	MATERIAL	ROCKWELL HARDNESS	PLATING
1.	Head Assembly	#544 Phosphor Bronze	----	Chrome
2.	Rear-end Housing	#544 Phosphor Bronze	----	Chrome
3.	Bearing Cap	#544 Phosphor Bronze	----	Chrome
4.	Slinger	AISI 303 Free Machine Stainless Steel	N/A	----

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

TABLE I. Materials, hardnessess and platings (Continued)

5.	Cover Ring	Acetyl Resin	N/A	----
6.	Pump Shaft	440 C Stainless (Annealed)	N/A	----
7.	Retaining Ring	303 Stainless Steel	N/A	----
8.	Pinion Gear	420F Stainless Steel	48-51	----
9.	Gear	420F Stainless Steel	48-51	----
10.	Gear Barrel	AISI 420 Stainless Steel	54-58	----
11.	Spindle Assembly	440C Stainless Steel	54-58	----

3.3 Design and dimensions. The handpiece shall be of the type known to the dental trade as a short sheath, round sleeve, right angle, prophylaxis handpiece. The handpiece shall be entirely suitable for use with round nose, belt driven and air driven, straight handpieces (see 6.1). The handpiece shall consist of a head assembly and back end assembly. The handpiece shall be provided with a screw stem socket threaded to receive (1-64 UNC) screw stem polishing brushes, 6520-00-890-1565, and (1-64 UNC) screw stem rubber cups, 6520-00-982-9377. The design and dimensions of the prophylaxis handpiece shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing No. 24488. The prophylaxis handpiece shall be capable of being autoclaved and lubricated without disassembly. The prophylaxis handpiece shall have the slinger ring press fitted to the gear barrel assembly to reduce penetration of prophylaxis pastes.

3.3.1 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier shall be responsible for preparing his own shop drawing. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:22Z
Check the source to verify that this is the current version before use.

3.4 Finish. The external surfaces of the handpiece shall be smooth. The finish shall be a satin finish except for the knurled portions.

3.5 Sterilization. The handpiece shall withstand a minimum of 62 steam autoclaves in a 14 day period, 5 cycles per day, without showing deterioration of components or breakdown of the handpiece. (See 4.4.7).

3.6 Runout. The runout of a true-running screw stem mandrel gage inserted in handpiece shall be not more than 0.006 inch, when tested as specified in 4.4.3.

3.7 Vibration and heat rise. The handpiece shall pass the test for vibration and heat rise as specified in 4.4.4.

3.8 Drive train smoothness and backlash. The handpiece shall show no binding, slipping, backlash, or irregular mechanical interference when tested as specified in 4.4.5.

3.9 End play. The end play of the screw stem portion of a rubber cup (see 3.2) inserted in the handpiece shall be no less than 0.0005 inch and no greater than 0.005 inch when tested as specified in 4.4.6.

3.10 Identification marking. Each handpiece shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Handpieces of foreign manufacture shall be marked with the name of the country of origin.

3.11 Service data. The contractor shall enclose, with each unit, two copies of each of one or more booklets which contain complete instructions for operation maintenance and all applicable sterilization methods.

3.12 Workmanship. The handpiece shall be free from defects which detracts from its appearance or impairs its serviceability.

3.13 Lubricating devices. Each unit shall be supplied with a lubricating nozzle and a bottle of lubricating grease.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:22Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials used in the handpieces, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III, and IV (see 4.2.1, 4.2.2, 4.3 and 4.3.1).

4.2.1 Sampling for examination. Sampling for examination for the handpiece shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in tables II. The unit of product for examination purposes shall be one handpiece.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:22Z
Check the source to verify that this is the currentversion before use.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (percent defective
Major A	II	1.0
Major B	S-1	2.5
Minor A	II	2.5
Minor B	S-1	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, for the handpiece. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one handpiece. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Inspection level
Material	3.1	4.4.1	1/
Hardness	3.1	4.4.2	S-1
Sterilization	3.4	4.4.7	S-1
Runout	3.5	4.4.3	S-1
Vibration and heat rise	3.6	4.4.4	S-1
Drive train smoothness and backlash	3.7	4.4.5	100%
End play	3.8	4.4.6	S-1

1/ A sufficient amount from each lot of raw materials used in the construction of the item shall be selected for chemical analysis when certificates of quality are not available as specified in 4.1.3.

4.3 Examination. The handpiece shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects.

Categories	Defects
Critical	None defined
Major A:	
101	Design and construction not as specified.
102	Prophylaxis handpiece does not fit and lock to round nose straight handpiece. (see 6.2).
103	Threads incomplete or stripped.
104	Polishing cup or brush when screwed in does not seat properly.
105	Handpiece shaft with and without polishing cup or brush does not turn freely and smoothly (binds, sticks, etc.)
Major B:	
151	Dimension affecting fit or handpiece to round nose straight handpiece not as specified.
152	Dimension affection affecting fit of prophylaxis cup or brush to head socket not as specified.
Minor A:	
201	Plating not free of blisters, or other defects that may cause peeling.
202	External surfaces of handpiece not smooth.
203	Finish not as specified.
204	Knurling not as specified, missing or incomplete.
Minor B:	
251	Dimension (other than 151 and 152) not as specified.
252	Service data or tools missing.
253	Identification marking not present, correct, complete, or legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

4.4.1 Chemical composition. In the absence of certificates of quality for raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4.3 Runout. A true-running test mandrel gage shall be screwed into socket of assembled handpiece and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy shall be employed. The difference between the maximum and minimum readings for one full rotation of the gage, measured 1/8 inch from screw socket, shall be reported as the runout.

4.4.4 Vibration and heat rise. The assembled and lubricated handpiece shall be tested by running the handpiece, containing a screw stem polishing brush, 6520-00-890-1565, for 3 minutes at 1500 ± 100 revolutions per minute against an acrylic disc under a 4 ± 1/10 ounce load. The handpiece shall be tested for excessive vibration when hand held; and excessive heat rise (above 105 degrees Fahrenheit) when tested with a pyrometer thermocuple fastened against the head housing and against the back end housing.

4.4.5 Drive train smoothness and backlash. The assembled and lubricated handpiece shall be tested by spinning the back end drive shaft through at least two full revolutions, both clockwise and counterclockwise, using finger pressure only as applied by the thumb and index finger. Any detectable binding, slipping, backlash, or irregular mechanical interference shall cause the handpiece to be rejected.

4.4.6 End play. A screw stem of a rubber cup shall be screwed into the socket of an assembled handpiece which is clamped securely in a vise. A dial indicator, sensitive to 0.0002 inch and equipped with a small ball tip shall be positioned on a flat surface of the screw stem perpendicular to the axis of the bur tube. The cup shall be pushed towards and away from the head of the handpiece so that end play can be measured. The difference between the maximum and minimum readings shall be reported as end play.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

4.4.7 Autoclaving. The handpiece shall be steam autoclaved at a temperature of 250-270°F (120-132°C) at 15-27 psi for 15-20 minutes. After the autoclaving the handpiece shall be:

1. Returned to room temperature.
2. Run at 3500 R.P.M. for 15 minutes.
3. Checked at head and neck for heat rise (105°F maximum temperature allowed).

One cycle constitutes the washing, steam autoclaving and lubrication of the handpiece.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each handpiece shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit Package. Each handpiece shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.1.4 Intermediate package. Ten handpieces shall be packaged in a box conforming to or PPP-B-636, type CF, class domestic. Closure shall be as specified in the box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for handpieces shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Ten handpieces shall be packaged in accordance with the requirements of MIl-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or c, as specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:22Z
Check the source to verify that this is the currentversion before use.

5.2.1 Level A. Five intermediate packages (50 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification, Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Five intermediate packages (50 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Five intermediate packages (50 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on 4- way entry pallets conforming to MIL-STD-147, Type IV or Type V. Pallet shall have a length of 40 inches and a width of 48 inches. Pallet loads, including the pallet, shall not exceed 43 inches in height, 43 inches in length and 52 inches in width. Quantity for shipment to one destination of less than 40 cubic feet need not be palletized.

5.2.5.1 Levels A and B. Handpieces packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Levels C. Handpieces packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6 NOTES

6.1 "Electrolizing" and "Duroplate" chromium alloy coatings are acceptable.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Tile, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

6.4 Subject term (key word) listing:
Handpiece
Prophylaxis
Right-Angle
Screw Stem

National Stock Number	Item Identification
6520-01-160-8382	HANDPIECE, DENTAL, Right Angle, Prophlaxis, Round Sleeve, Screw Stem

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air-Force-03

Review Activities:

Army-MD
Navy-MS
Air-Force-03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

51798

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1871

Location: NCR Disk #8

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-ATTS
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-36809C	HANDPIECE, DENTAL, RIGHT-ANGLE, PROPHYLAXIS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark on*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Inclu Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:22Z
Check the source to verify that this is the current version before use.